

Yi Hu

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EDUCATION

PhD in Biomedical Engineer

University of Alberta

Jan. 2020 – (Expected May 2026)

Edmonton, Canada

MSc in Control Theory and Control Engineering

Sichuan University

Sept. 2017 – Jun. 2020

Chengdu, China

B.S. in Process Equipment and Control Engineering

Hefei University of Technology

Sept. 2013 – Jun. 2027

Hefei, China

PERSONAL SUMMARY

Robotics engineer with a PhD in autonomous surgical systems, specializing in trajectory generation, motion optimization, and data-driven learning for robotic ultrasound and endoscopic scanning. Experienced in integrating perception, planning, and policy learning for high-precision autonomous manipulation.

RESEARCH EXPERIENCE

Human-Centered Autonomy Lab

Graduate Research Assistant, University of Alberta

Jan. 2024 – Present

Edmonton, Canada

- Integrated a 3D spatial representation module into VLA models (**Pi0**, **OpenVLA**) to enable cross-modal fusion and 3D-aware manipulation; developed a Mixture-of-Experts action head for improved geometric reasoning. Fine-tuned large vision-language-action (VLA) models (Pi0) on real-world robot datasets using NVIDIA A6000 GPUs; deployed on a Franka robot with real-time RGB-D image inference.
- Develop a single-step diffusion framework for temporally structured motion primitives. Achieved 7–10× faster inference on MetaWorld and ManiSkills. Validated on real-world Franka robot in a high-speed ball catching task.
- Designed and implemented a cross-embodiment reinforcement-learning framework to transfer human daily-activity demonstrations to robotic surgical tasks; validated on the da Vinci robot for precise tool manipulation and camera-view control.

Telerobotic and Biorobotic Systems Group

Graduate Research Assistant, University of Alberta

Jan. 2021 – Dec. 2023

Edmonton, Canada

- Developed an autonomous ultrasound scanning framework on the Franka robot, integrating **admittance control** and tissue-probe force contact regulation; trained and deployed a **U-Net** model for ultrasound image assessment.
- Implemented a **hand-pose-based teleoperation system** for 6-DoF ultrasound probe manipulation; built a vision pipeline for 3D hand pose estimation and mapped human motion to robot end-effector with safety constraints.
- Implemented a **surgical view control** system with Endoscopic Camera Manipulator on da Vinci robot; applied unsupervised motion primitive learning to optimize camera trajectories in constrained surgical workspaces.
- Developed precise, collision-aware 6-DoF manipulation with Patient Side Manipulator on da Vinci robot; developed smooth trajectory generation and orientation-constrained motion planning algorithms.

Perception, Motion Control and Intelligent Robot Innovation Lab

Graduate Research Assistant, Sichuan University

Sept. 2017 – Dec. 2020

Chengdu, China

- Built an event-camera (DVS) SLAM pipeline for a wheeled inspection robot using a front-mounted camera. Fused event vision with proprioceptive sensors (wheel odometry / IMU) to improve robustness during long-horizon navigation.
- Built a Unity simulation environment for a magnetic wheeled mobile robot. Modeled magnetic adhesion and vehicle dynamics to enable stable motion on vertical and curved container walls.
- Designed and simulated control algorithms for **fixable-joint manipulators** under actuator saturation and external disturbances, improving trajectory stability and robustness.
- Modeled and controlled a **magnetic-wheel mobile robot**, deriving kinematic and dynamic models for precise trajectory tracking.
- Developed a Unity-based simulation environment for magnetic-wheel robot navigation and visual localization.

TECHNICAL SKILLS

Robots: da Vinci Robot, Franka Robot, Magnetic Mobile Vehicle, PHANToM Premium 1.5A Robot

Languages: English, Chinese, Cantonese